

OSDMamba: Enhancing Oil Spill Detection From Remote Sensing Images Using Selective State-Space Model

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Abstract

This paper proposes OSDMamba, a novel semantic segmentation framework for oil spill detection in remote sensing images based on selective state-space models. Unlike conventional convolutional neural networks (CNNs) that suffer from limited receptive fields, OSDMamba leverages the Mamba architecture to efficiently capture long-range contextual dependencies. An asymmetric decoder integrating convolutional state-space models and deep supervision is introduced to address class imbalance and enhance multiscale feature fusion. Experimental results on two public datasets demonstrate that OSDMamba achieves state-of-the-art performance.

1 Introduction

Marine oil spills pose serious threats to marine ecosystems and coastal environments. Rapid and accurate oil spill detection is critical for environmental protection and emergency response. With the increasing availability of optical and SAR remote sensing imagery, deep learning has become a promising tool for automatic oil spill detection.

Most existing oil spill detection methods rely on CNN-based encoder-decoder architectures. While effective in capturing local features, CNNs have limited receptive fields and struggle to detect small oil spill regions. Moreover, oil spills are rare events, leading to severe class imbalance where oil spill pixels occupy only a small portion of the image.

Transformer-based models address global context modeling but suffer from high computational complexity and excessive feature smoothing, making them less suitable for oil spill detection tasks.

To overcome these challenges, this paper explores state-space models (SSMs), specifically the Mamba architecture, and proposes OSDMamba as the first Mamba-based framework tailored for oil spill detection.

2 Related Work

Traditional oil spill detection methods include thresholding, texture analysis, and classical machine learning techniques. These approaches are sensitive to noise and environmental variability.

CNN-based semantic segmentation models such as U-Net and its variants significantly improve detection accuracy but are limited by local receptive fields. Transformer-based approaches model global dependencies but incur high computational cost and may miss fine details.

Recent advances in state-space models demonstrate strong capability in capturing long-range dependencies with linear complexity. However, their application to oil spill detection has not been previously explored.

3 Proposed OSDMamba Framework

OSDMamba follows an encoder-decoder architecture. The encoder is based on a modified Vision Mamba (VMamba) model, while the decoder adopts an asymmetric design with ConvSSM modules and deep supervision.

The framework is designed to expand the effective receptive field, preserve minority class features, and improve segmentation accuracy under class imbalance.

4 Encoder Design

4.1 Vision State-Space Block

The Vision State-Space (VSS) block is the core component of the encoder. To process two-dimensional image data, a 2-D selective scanning (SS2D) mechanism is employed. The input feature map is unfolded along four directions, processed using state-space models, and merged back into a 2-D representation.

This bidirectional scanning strategy enables effective modeling of both local and global contextual information while maintaining computational efficiency.

4.2 Encoder Structure

The encoder follows a multistage design with progressive downsampling. The number of VSS blocks in the four stages is configured as $\{2, 2, 9, 2\}$, balancing feature extraction capacity and computational cost. ImageNet-pretrained weights are used to accelerate convergence and improve generalization.

5 Decoder Design

5.1 Convolutional State-Space Model

To combine local spatial modeling with global sequence modeling, convolutional state-space models (ConvSSM) are introduced. ConvSSM integrates convolution operations with state-space dynamics, enabling efficient feature extraction while preserving global context.

5.2 Asymmetric Decoder Structure

The decoder adopts an asymmetric design. In the early decoding stages, VSS blocks and ConvSSM modules are used to enhance detection of small oil spill regions. In later stages, lightweight upsampling operations are applied to reduce computational cost.

Deep supervision is applied at multiple scales using auxiliary outputs, guiding feature learning and improving segmentation performance for minority classes.

6 Datasets

Experiments are conducted on two publicly available datasets:

- The M4D dataset, containing SAR images with five semantic categories including oil spill and look-alikes.
- The MADOS dataset, a large-scale benchmark consisting of multispectral Sentinel-2 images with diverse marine pollution categories.

Both datasets exhibit strong class imbalance, making them suitable for evaluating the effectiveness of OSDMamba.

7 Implementation Details

The model is implemented using PyTorch and trained on an NVIDIA GPU. AdamW optimization is used with pretrained ImageNet weights. A hybrid loss function combining focal loss and Dice loss is adopted to address class imbalance and improve boundary accuracy.

Training is conducted for 100 epochs with data augmentation and multiscale supervision.

8 Methodology

The complete detection methodology can be summarized as follows:

1. Divide input remote sensing images into patches.
2. Extract multiscale features using the VMamba-based encoder.
3. Model long-range dependencies using selective state-space scanning.
4. Fuse multiscale features using an asymmetric decoder with ConvSSM.
5. Apply deep supervision to enhance learning of minority classes.
6. Generate pixel-level oil spill segmentation maps.

This methodology effectively addresses limited receptive fields and class imbalance.

9 Performance Analysis

Quantitative evaluations demonstrate that OSDMamba significantly outperforms state-of-the-art CNN- and Transformer-based methods. On the M4D dataset, OSDMamba achieves a mean Intersection over Union (mIoU) of 70.25%, with a 12.18% improvement in the oil spill category compared to U-Net.

On the MADOS dataset, OSDMamba improves mIoU by 8.9% and F1-score by 0.6% over competing methods. Ablation studies confirm that the asymmetric decoder and deep supervision play critical roles in performance improvement.

False positive analysis further shows that OSDMamba effectively reduces misclassification in minority classes, demonstrating robustness under severe class imbalance.

10 Conclusion

This paper introduces OSDMamba, the first Mamba-based architecture for oil spill detection in remote sensing images. By leveraging selective state-space modeling, an asymmetric decoder, and deep supervision, the proposed framework effectively captures global context while preserving fine details.

Extensive experiments confirm that OSDMamba achieves state-of-the-art performance on multiple datasets. The proposed approach offers a new direction for oil spill detection and other remote sensing segmentation tasks involving rare and small targets.